

Overview

Probability and statistics are core to AI because they model uncertainty, risk, and data variability. The Titanic dataset is a practical case for probability estimation and descriptive analysis.

Core Probability Concepts

- Random variables and events.
- Marginal, joint, and conditional probability.
- Independence vs dependence.

Core Statistics Concepts

- Mean, median, variance, and standard deviation.
- Percentiles and quantiles.
- High-level distributions: Bernoulli, binomial, normal.

Probability & Statistics on Titanic

- Compute baseline survival probability $P(\text{survived})$.
- Compute conditional probabilities such as $P(\text{survived} \mid \text{sex=female})$ and $P(\text{survived} \mid \text{class})$.
- Compare age/fare distributions between groups.

Common Pitfalls

- Class imbalance can bias naive interpretations.
- Correlation is not causation.
- Conditional probabilities are often misread without context.

Business Use Cases

- Risk scoring and prioritization.
- Resource allocation under uncertainty.
- Decision support with probabilistic evidence.

Interview Prep Checklist

- Explain conditional probability with a business example.
- Interpret survival probabilities and group-level differences.
- Discuss implications of class imbalance in evaluation.